

Compound Interest Calculator

Canada · CAD (CA\$)

Personal Financial Scenario Report

fincalcsmart.com/compound-interest-calculator

EXECUTIVE SUMMARY

Good

CA\$108,959

Est. Future Value
20 years projection

CA\$53,000

Total Contributions

CA\$55,959

Est. Investment Growth
51.4% of balance

77/100

Compounding Power
Good

CONTRIBUTIONS VS. ESTIMATED INVESTMENT GROWTH

49%

51%

Total projected: CA\$108,959

Contributions: CA\$53,000 (48.6%) Estimated Investment Growth: CA\$55,959 (51.4%)

AI-ASSISTED GROWTH SUMMARY

Based on a 6% assumed annual return with monthly compounding over 20 years, this scenario projects an estimated future value of CA\$108,959. Investment growth accounts for 51.4% of the total, reflecting a meaningful compounding effect over the chosen time horizon.

Contributions of CA\$53,000 represent 49% of the projected balance. Estimated investment growth of CA\$55,959 accounts for the remaining 51% — assuming the 6% rate remains constant for the full 20 years. These projections do not account for taxes, inflation, or investment fees.

Adding CA\$100/month to regular contributions would increase the projected balance by approximately CA\$46,204, reaching CA\$155,163 after 20 years. Consistent contributions amplified by compounding time are the primary drivers of long-term investment growth.

INPUTS & ASSUMPTIONS

Initial Investment	CA\$5,000
Monthly Contribution	CA\$200.00/month
Assumed Annual Return	6%
Compound Frequency	Monthly (12×/year)
Investment Horizon	20 years
Return Type	Nominal (pre-inflation, before fees and taxes)

SCENARIO BREAKDOWN

Initial Investment	CA\$5,000
Monthly Contribution	CA\$200.00/month
Investment Horizon	20 years
Total Contributions	CA\$53,000
Estimated Investment Growth	CA\$55,959
Estimated Future Value	CA\$108,959
Contribution Share	49%
Growth Share	51%
Compounding Power Score	77/100 (Good)

KEY DRIVERS / POSSIBLE ADJUSTMENTS

- Compounding Power Score: 77/100 (Good). At 51.4% of the final balance attributed to investment growth, compounding is working effectively over this 20-year horizon. The effect accelerates non-linearly — the final years contribute disproportionately more than the first.
- Projected balance milestones at the current rate: CA\$41,873 at 10 years and CA\$231,016 at 30 years. The gap between each 10-year interval typically grows larger — a sign that compounding acceleration increases with time.
- Contribution potential: adding CA\$100/month to regular contributions would increase the projected balance by approximately CA\$46,204, reaching CA\$155,163 after 20 years. This extra amount benefits from the full remaining compounding period.

METHODOLOGY

What this calculator models:

- Converts the nominal annual return rate to an Effective Annual Rate (EAR) using the selected compounding frequency, then derives an Effective Monthly Rate (EMR) for consistent period-to-period calculations.
- Applies the standard future-value annuity formula to calculate the future value of the initial investment and ongoing monthly contributions at the derived effective monthly rate.
- Computes the Compounding Power Score from the percentage of the projected future value attributable to investment growth versus total contributions; a higher score indicates compounding is playing a larger role.
- Projects balance milestones at 10, 20, and 30 years using the same rate and contribution inputs to illustrate the non-linear acceleration of compounding over time.

What this calculator does not model:

- Taxes on investment gains, withholding taxes, or tax wrapper benefits (e.g. TFSA, RRSP, 401(k), ISA).
- Investment fees, management expense ratios (MER), fund charges, or advisory costs that would reduce net returns.
- Inflation — all projected values are nominal and are not adjusted for changes in purchasing power. A nominal return rate is not the same as a real (inflation-adjusted) return.

EDUCATIONAL DISCLAIMER

This report is for illustrative and educational purposes only. Projected future values are estimates based on a fixed nominal annual return rate, regular monthly contributions, and the selected compounding frequency. Actual investment returns vary and are not guaranteed — past performance does not predict future results. This model does not account for taxes, inflation, investment fees, market volatility, early withdrawals, or changes in contribution amounts over time. The Compounding Power Score reflects the proportion of the projected balance attributable to investment growth under the stated assumptions; it does not represent a guarantee or prediction of investment performance. Results do not constitute financial, investment, tax, or legal advice. Consult a qualified financial advisor before making any investment or savings decisions.